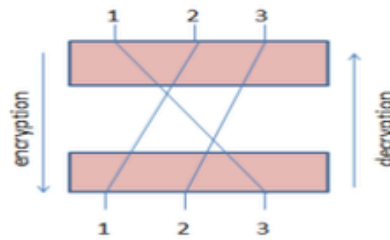


Question 1:

A. Encrypt the following message using transposition cipher based on the key shown below:



The message is "SECRET WRITING".ZZ

Answer: ECSETRATWTINTZZG

Using the same key, decrypt the cipher text: MASLBLATO

Answer: SMALLBOAT

$$Q = 16 + 6 = 22 \pmod{26} = 22 \text{ W}$$

$$U = 21 + 6 = 27 \pmod{26} = 1 \text{ A}$$

$$I = 8 + 6 = 14 \pmod{26} = 14 \text{ O}$$

$$Z = 25 + 6 = 31 \pmod{26} = 5 \text{ F}$$

$$S = 18 - 6 = 12 \pmod{26} = 12 \text{ M}$$

$$G = 6 - 6 = 0 \pmod{26} = 0 \text{ A}$$

$$Y = 24 - 6 = 18 \pmod{26} = 18 \text{ S}$$

$$Z = 25 - 6 = 19 \pmod{26} = 19 \text{ T}$$

$$K = 10 - 6 = 4 \pmod{26} = 4 \text{ E}$$

$$X = 23 - 6 = 17 \pmod{26} = 17 \text{ R}$$

B. Alice wants to encrypt a message to Bob by using the Caesar cipher algorithm and the key=6.

The plaintext = "QUIZ".

Answer: WAOF

Using the same key, decrypt the message: SGYZKX

Answer: MASTER

Question 2:

The following ciphertext was generated using a simple substitution algorithm:

53†††305)) 6*;4826) 4†.) 4†; ; 806*;48†8¶60)) 85;; ;] 8*; ;†*8†83
(88) 5*†;46 (;88*96*?;8) *† (;485) ; 5*†2: *† (;4956*2 (5*-4) 8¶8*
;4069285) ;) 6†8) 4††;1 (†9;48081;8:8†1;48†85;4) 485†528806*81
(†9;48; (88;4 (†?34;48) 4†;161; :188;†?;

Symbolic = number of repetitions

4 = 16	† = 15	? = 3
: = 4	3 = 4	q = 6
; = 27	5 = 12	(= 9
† = 14	. = 1	j = 1
6 = 11	1 = 5	‡ = 2
) = 16	8 = 34	
0 = 6	1 = 6	
† = 8	- = 1	

1. In English, the most frequently occurring letter is E. Also, e is often appears in pairs (e.g., meet, speed, seen, been, agree, etc.). Try to find a character in the ciphertext that decodes to E.

Answer: 8

2. The most common word in English is "the." Use this fact to guess the characters that stand for t and h.

Answer: ; 48

3. Is this encryption algorithm classified as monoalphabetic cipher or polyalphabetic cipher? And how did that help you decrypting the text?

Answer: monoalphabetic because every character have a one character in ciphertext.

Question 3:

A. In DES, if $M = 6000\ 3000\ 8000\ 0000$ (Hex), Find the IP(M).

Question 3:

A. In DES, if $M = 6000\ 3000\ 8000\ 0000$ (Hex), Find the $IP(M)$.

Original M

M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8
M_9	M_{10}	M_{11}	M_{12}	M_{13}	M_{14}	M_{15}	M_{16}
M_{17}	M_{18}	M_{19}	M_{20}	M_{21}	M_{22}	M_{23}	M_{24}
M_{25}	M_{26}	M_{27}	M_{28}	M_{29}	M_{30}	M_{31}	M_{32}
M_{33}	M_{34}	M_{35}	M_{36}	M_{37}	M_{38}	M_{39}	M_{40}
M_{41}	M_{42}	M_{43}	M_{44}	M_{45}	M_{46}	M_{47}	M_{48}
M_{49}	M_{50}	M_{51}	M_{52}	M_{53}	M_{54}	M_{55}	M_{56}
M_{57}	M_{58}	M_{59}	M_{60}	M_{61}	M_{62}	M_{63}	M_{64}

$IP(M)$

M_{58}	M_{50}	M_{42}	M_{34}	M_{26}	M_{18}	M_{10}	M_2
M_{60}	M_{52}	M_{44}	M_{36}	M_{28}	M_{20}	M_{12}	M_4
M_{62}	M_{54}	M_{46}	M_{38}	M_{30}	M_{22}	M_{14}	M_6
M_{64}	M_{56}	M_{48}	M_{40}	M_{32}	M_{24}	M_{16}	M_8
M_{57}	M_{49}	M_{41}	M_{33}	M_{25}	M_{17}	M_9	M_1
M_{59}	M_{51}	M_{43}	M_{35}	M_{27}	M_{19}	M_{11}	M_3
M_{61}	M_{53}	M_{45}	M_{37}	M_{29}	M_{21}	M_{13}	M_5
M_{63}	M_{55}	M_{47}	M_{39}	M_{31}	M_{23}	M_{15}	M_7

IP(M) = 0104 0000 1005 0000 (Hex)

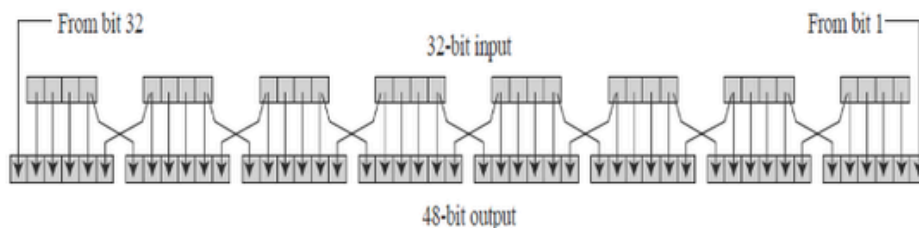
B. Using the S-box shown below, What is the output if the input was: 011011 and 101001

Inner 4 bits																
	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
00	0010	1100	0100	0001	0111	1100	1011	0110	1000	0101	0011	1111	1101	0000	1110	1001
01	1110	1011	0010	1100	0100	0111	1101	0001	0101	0000	1111	1100	0011	1001	1000	0110
10	0100	0010	0001	1011	1100	1101	0111	1000	1111	1001	1100	0101	0110	0011	0000	1110
11	1011	1000	1100	0111	0001	1110	0010	1101	0110	1111	0000	1001	1100	0100	0101	0011

Answer: 1) 1001 2) 0001

C. In the first step in the function F of the DES encryption, the 32-bit right side is expanded in an expansion permutation to get a 48-bit output. Use the expansion permutation below on the message :

R2 = 0000 1111 0101 1010 1111 0000 1010 1000



Answer (binary): 000001 011110 101001 110101 011110 100001 01011 010000

D. In the step of creating the sub key for the 4th round, C3 and D3 will be shifted to the left. Find the result of C4 and D4.

$C_3 = 10011001010101111000011111$
 $D_3 = 001111101010101110011111$

$C_4 = 0011001010101011110000111111$
 $D_4 = 011110100101011001111100$

Question 4:

Using this Playfair matrix:

M	F	H	I/J	K
U	N	O	P	Q
Z	V	W	X	Y
E	L	A	R	G
D	S	T	B	C

Mu
u z
 St
T B
 Se
D L
 eJ
G Z
 ou
P N
 ov
N W
 eY
L G
 Ca
T G
 do
T U
 ga
E A
 hw
O V
 es
L O
 tc
B D
 om
U H
 in
F P
 ga
E A
 to
H W
 nc
Q S
 ez
D E

- a. Encrypt this message: Must see you over Cadogan West. Coming at once.
 b. Construct a Playfair matrix with the key largest.
 c. Construct a Playfair matrix with the key occurrence.

a) U Z T B D L G Z P N N W L G T G T U E A O V L D B D U H F P E A H W Q S D E

b)

L	A	R	G	E
S	T	B	C	D
F	H	I/J	K	M
N	O	P	Q	U
V	W	X	Y	Z

c)

O	C	U	R	E
N	A	B	D	F
G	H	I/J	K	L
M	P	Q	S	T
V	W	X	Y	Z